

Definition

A sequence of real numbers or a sequence in $\mathbb{R}$ is a function $f : \mathbb{N} \rightarrow \mathbb{R}$.

- We write a_n for $f(n)$, $n \in \mathbb{N}$ and the notation for a sequence is $\{a_n\}_{n=1}^{\infty}$.

Examples

- 1 Constant sequence: $\{c, c, c, \dots\}$, where $c \in \mathbb{R}$.
- 2 Sequence defined by listing: $\{1, 4, 8, 11, 52, \dots\}$.
- 3 Sequence defined by rule: $\{a_n\}_{n=1}^{\infty}$, where $a_n = 3n^2$ for all $n \in \mathbb{N}$.
- 4 $\left\{ \frac{(-1)^{n+1}}{n} \right\}_{n=1}^{\infty}$
- 5 $\left\{ \frac{n-1}{n} \right\}_{n=1}^{\infty}$
- 6 $\{\sqrt{n}\}_{n=1}^{\infty}$

- What does **convergence** mean?
- Think of the examples: $\{2, 2, 2, \dots\}$, $\{\frac{1}{n}\}_{n=1}^{\infty}$, $\{n^2 - 1\}_{n=1}^{\infty}$, $\{1, 2, 1, 2, \dots\}$, $\{(-1)^n \frac{1}{n}\}_{n=1}^{\infty}$, $\{(-1)^n (1 - \frac{1}{n})\}_{n=1}^{\infty}$.

Definition

A sequence $\{a_n\}_{n=1}^{\infty}$ **converges** to limit L if for every $\epsilon > 0$ (given) there exists a positive integer N such that $n \geq N \implies |a_n - L| < \epsilon$.

- Notation: $L = \lim_{n \rightarrow \infty} a_n$ or $a_n \rightarrow L$.
- If $\{a_n\}_{n=1}^{\infty}$ is a sequence and if both $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} a_n = M$ holds, then $L = M$.

Examples

- 1 Constant sequence $\{c\}_{n=1}^{\infty}$, $c \in \mathbb{R}$, has c as its limit.
- 2 Show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Solution: Let $\epsilon > 0$ be given. To show that $1/n$ approaches 0, we must show that there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \epsilon.$$

But $1/n < \epsilon \Leftrightarrow n > 1/\epsilon$. Thus, if we choose $N \in \mathbb{N}$ such that $N > 1/\epsilon$, then for all $n \geq N$, $1/n < \epsilon$.

Example

Show that $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$.

Solution: For any $\epsilon > 0$,

$$\left| \frac{(-1)^n}{n} - 0 \right| = \frac{1}{n} < \epsilon \quad \forall n \geq N,$$

where N is a positive integer such that $N > \frac{1}{\epsilon}$. Thus, $\frac{(-1)^n}{n} \rightarrow 0$ as $n \rightarrow \infty$.

Example

Show that $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$.

Solution: Note that $|a_n - 1| = \frac{1}{n+1} < \frac{1}{n}$. Thus, for any $\epsilon > 0$, take $N > \frac{1}{\epsilon}$, we get

$$\left| \frac{n}{n+1} - 1 \right| = \frac{1}{n+1} < \frac{1}{n} < \epsilon \quad \forall n \geq N.$$

Hence, $\frac{n}{n+1} \rightarrow 1$ as $n \rightarrow \infty$.

Theorem

Let $\{a_n\}_1^\infty$ and $\{b_n\}_1^\infty$ be two sequences such that $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$. Then

- (i) $\lim_{n \rightarrow \infty} (a_n + b_n) = L + M$.
- (ii) $\lim_{n \rightarrow \infty} (a_n - b_n) = L - M$.
- (iii) $\lim_{n \rightarrow \infty} (ca_n) = cL, \quad c \in \mathbb{R}$.
- (iv) $\lim_{n \rightarrow \infty} (a_nb_n) = LM$.
- (v) $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{L}{M}$ if $M \neq 0$.

Examples

Find the limit of the following sequences:

- (i) $\left\{ \frac{5}{n^2} \right\}_1^\infty$, (ii) $\left\{ \frac{3n^2 - 6n}{5n^2 + 4} \right\}_1^\infty$, (iii) $\lim_{n \rightarrow \infty} \left(\frac{n-1}{n} \right)$.

Solution: (i)

$$\lim_{n \rightarrow \infty} \frac{5}{n^2} = \lim_{n \rightarrow \infty} 5 \cdot \frac{1}{n} \cdot \frac{1}{n} = 5 \cdot \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} = 5 \cdot 0 \cdot 0 = 0.$$

(ii) Notice that

$$\frac{3n^2 - 6n}{5n^2 + 4} = \frac{3 - 6/n}{5 + 4/n^2}.$$

Now

$$\lim_{n \rightarrow \infty} (3 - 6/n) = 3 - 6 \lim_{n \rightarrow \infty} 1/n = 3 - 6 \cdot 0 = 3$$

and

$$\lim_{n \rightarrow \infty} (5 + 4/n^2) = 5 + 4 \lim_{n \rightarrow \infty} 1/n^2 = 5 + 4 \cdot 0 = 5.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{3n^2 - 6n}{5n^2 + 4} = \lim_{n \rightarrow \infty} \frac{3 - 6/n}{5 + 4/n^2} = \frac{3}{5}.$$

(iii)

$$\lim_{n \rightarrow \infty} \left(\frac{n-1}{n} \right) = \lim_{n \rightarrow \infty} \frac{1 - 1/n}{1} = 1 - \lim_{n \rightarrow \infty} \frac{1}{n} = 1 - 0 = 1.$$

Sandwich theorem for sequences

Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be three sequences such that $a_n \leq b_n \leq c_n$ for all $n \in \mathbb{N}$. If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Proof: Let $\epsilon > 0$ be given. As $\lim_{n \rightarrow \infty} a_n = L$, there exists $N_1 \in \mathbb{N}$ such that

$$n \geq N_1 \implies |a_n - L| < \epsilon. \quad (1)$$

Similarly as $\lim_{n \rightarrow \infty} c_n = L$, there exists $N_2 \in \mathbb{N}$

$$n \geq N_2 \implies |c_n - L| < \epsilon. \quad (2)$$

Let $N = \max\{N_1, N_2\}$. Then, $L - \epsilon < a_n$ (from (1)) and $c_n < L + \epsilon$ (from (2)). Thus

$$L - \epsilon < a_n \leq b_n \leq c_n < L + \epsilon.$$

Thus $|b_n - L| < \epsilon$ for all $n \geq N$. Hence the proof.

Examples

Using Sandwich theorem, prove the following:

- (i) $\lim_{n \rightarrow \infty} \frac{\cos n}{n} = 0$.
- (ii) $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$.
- (iii) $\lim_{n \rightarrow \infty} (-1)^n \frac{1}{n} = 0$.
- (iv) If $0 < b < 1$, then $\lim_{n \rightarrow \infty} b^n = 0$.
- (v) $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.

Solution: (i) Consider the sequence $\left\{ \frac{\cos n}{n} \right\}_{n=1}^{\infty}$. Then $\frac{-1}{n} \leq \frac{\cos n}{n} \leq \frac{1}{n}$. Hence by Sandwich theorem, $\lim_{n \rightarrow \infty} \frac{\cos n}{n} = 0$.

(ii) As $0 < \frac{1}{2^n} < \frac{1}{n}$ and $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, $\frac{1}{2^n}$ also converges to 0 by Sandwich theorem.

(iii) As $\frac{-1}{n} \leq (-1)^n \frac{1}{n} \leq \frac{1}{n}$ for all $n \geq 1$ and $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$, $(-1)^n \frac{1}{n}$ also converges to 0 by Sandwich theorem.

(iv) Since $0 < b < 1$, we can write $b = \frac{1}{1+a}$, where $a := \frac{1}{b} - 1$ so that $a > 0$. Also we have $(1+a)^n \geq 1+na$. Hence

$$0 < b^n = \frac{1}{(1+a)^n} \leq \frac{1}{1+na} < \frac{1}{na}.$$

So, by sandwich Theorem, we conclude that $\lim_{n \rightarrow \infty} b^n = 0$.

(v) Let $a_n = n^{\frac{1}{n}} - 1$. Then $0 \leq a_n < 1$ for all $n \in \mathbb{N}$. Further,

$$n = (1 + a_n)^n \geq \frac{n(n-1)}{2} a_n^2.$$

Thus $0 \leq a_n \leq \sqrt{\frac{2}{(n-1)}}$ ($n \geq 2$). As $\sqrt{\frac{2}{(n-1)}} \rightarrow 0$ as $n \rightarrow \infty$, by Sandwich theorem, $a_n \rightarrow 0$, i.e., $n^{\frac{1}{n}} \rightarrow 1$ as $n \rightarrow \infty$.